# § 2.4

**In exercises 1–16, find the derivative of each function.**

1. $f(x) = (x^2 + 3)(x^3 - 3x + 1)$
2. $f(x) = (x^3 - 2x^2 + 5)(x^4 - 3x^2 + 2)$
3. $f(x) = (\sqrt{x} + 3x)\left(5x^2 - \frac{3}{x}\right)$
4. $f(x) = (x^{3/2} - 4x)\left(x^4 - \frac{3}{x^2} + 2\right)$

5. $g(t) = \frac{3t - 2}{5t + 1}$
6. $g(t) = \frac{t^2 + 2t + 5}{t^2 - 5t + 1}$
7. $f(x) = \frac{3x - 6\sqrt{x}}{5x^2 - 2}$
8. $f(x) = \frac{6x - 2/x}{x^2 + \sqrt{x}}$
9. $f(u) = \frac{(u + 1)(u - 2)}{u^2 - 5u + 1}$
10. $f(x) = \frac{2x}{x^2 + 1}(x + 3)$
11. $f(x) = \frac{x^2 + 3x - 2}{\sqrt{x}}$
12. $f(u) = \frac{u^2 - 2u}{u^2 + 5u}$
13. $h(t) = t(\sqrt[3]{t} + 3)$
14. $h(t) = \frac{t^2}{3} + \frac{5}{t^2}$
15. $f(x) = (x^2 - 1)\frac{x^3 + 3x^2}{x^2 + 2}$
16. $f(x) = (x + 2)\frac{x^2 - 1}{x^2 + x}$

$-3x^{-2}$

4. $f(x) = \left(x^{\frac{3}{2}} - 4x\right)\left(x^4 - \frac{3}{x^2} + 2\right)$

$= \left(\frac{3}{2}x^{\frac{1}{2}} - 4\right)\left(x^4 - \frac{3}{x^2} + 2\right)$

$+ \left(x^{\frac{3}{2}} - 4x\right)\left(4x^3 + \frac{6}{x^3}\right)$

$g \quad (g^{-1})' = -1g^{-2} \cdot g'$

8. $f(x) = \frac{6x - 2x^{-1}}{x^2 + x^{\frac{1}{2}}} = (6x - 2x^{-1})\left(x^2 + x^{\frac{1}{2}}\right)^{-1}$

$f'(x) = (6 + 2x^{-2})\left(x^2 + x^{\frac{1}{2}}\right)^{-1} + (6x - 2x^{-1})(-1) \cdot \left(x^2 + x^{\frac{1}{2}}\right)^{-2} \cdot \left(2x + \frac{1}{2}x^{-\frac{1}{2}}\right)$

$= (6 + 2x^{-2})\left(x^2 + x^{\frac{1}{2}}\right)^{-1} - (6x - 2x^{-1})\left(x^2 + x^{\frac{1}{2}}\right)^{-2}\left(2x + \frac{1}{2}x^{-\frac{1}{2}}\right)$

15. $f(x) = (x^2 - 1)\frac{x^3 + 3x^2}{x^2 + 2} = (x^2 - 1)(x^3 + 3x^2) \cdot (x^2 + 2)^{-1}$

$f'(x) = 2x(x^3 + 3x^2) \cdot (x^2 + 2)^{-1} + (x^2 - 1)(3x^2 + 6x)(x^2 + 2)^{-1}$

$+ (x^2 - 1)(x^3 + 3x^2) \cdot (-1)(x^2 + 2)^{-2} \cdot (2x)$

**In exercises 21–24, assume that $f$ and $g$ are differentiable with $f(0) = -1$, $f(1) = -2$, $f'(0) = -1$, $f'(1) = 3$, $g(0) = 3$, $g(1) = 1$, $g'(0) = -1$ and $g'(1) = -2$. Find an equation of the tangent line to the graph of $y = h(x)$ at $x = a$.**

**21.** $h(x) = f(x)g(x)$ at (a) $a = 0$, (b) $a = 1$.

**22.** $h(x) = \dfrac{f(x)}{g(x)}$ at (a) $a = 1$, (b) $a = 0$.

**23.** $h(x) = x^2 f(x)$ at (a) $a = 1$, (b) $a = 0$.

**24.** $h(x) = \dfrac{x^2}{g(x)}$ at (a) $a = 1$, (b) $a = 0$.

$f(1) = -2$

$f'(1) = 3$

$g(1) = 1$

$g'(1) = -2$

24. $h'(1) = ?$

$$h'(x) = \frac{2x}{g(x)} - \frac{x^2 g'(x)}{(g(x))^2}$$

$$h'(1) = \frac{2}{g(1)} - \frac{g'(1)}{(g(1))^2}$$

$$= \frac{2}{1} - \frac{-2}{1} = 2 + 2 = 4$$

§2.5

**In exercises 5–16, differentiate each function.**

**5.** (a) $f(x) = (x^3 - x)^3$ (b) $f(x) = \sqrt{x^2 + 4}$

**6.** (a) $f(x) = (x^3 + x - 1)^3$ (b) $f(x) = \sqrt{4x - 1/x}$

**7.** (a) $f(t) = t^5\sqrt{t^3 + 2}$ (b) $f(t) = (t^3 + 2)\sqrt{t}$

**8.** (a) $f(t) = (t^4 + 2)\sqrt{t^2 + 1}$ (b) $f(t) = \sqrt{t}(t^{4/3} + 3)$

**9.** (a) $f(u) = \dfrac{u^2 + 1}{u + 4}$ (b) $f(w) = \dfrac{w^3}{(w^2 + 4)^2}$

**10.** (a) $f(v) = \dfrac{v^2 - 1}{v^2 + 1}$ (b) $f(x) = \dfrac{x^2 + 4}{(x^3)^2}$

**11.** (a) $g(x) = \dfrac{x}{\sqrt{x^2 + 1}}$ (b) $h(x) = \sqrt{\dfrac{x}{x^2 + 1}}$

**12.** (a) $g(x) = x^2\sqrt{x + 1}$ (b) $h(x) = \sqrt{(x^2 + 1)(\sqrt{x} + 1)^3}$

**13.** (a) $h(x) = \dfrac{6}{\sqrt{x^2 + 4}}$ (b) $h(x) = \dfrac{\sqrt{x^2 + 4}}{6}$

**14.** (a) $h(t) = \dfrac{(t^3 + 4)^5}{8}$ (b) $h(t) = \dfrac{8}{(t^3 + 4)^5}$

**15.** (a) $f(x) = (\sqrt{x^3 + 2} + 2x)^{-2}$ (b) $f(x) = \sqrt{x^3 + 2 + 2x^{-2}}$

**16.** (a) $f(x) = \sqrt{4x^2 + (8 - x^2)^2}$ (b) $f(x) = (\sqrt{4x^2 + 8} - x^2)^2$

**In exercises 17–20, name the method (chain rule, product rule, quotient rule) that you would use first to find the derivative of the function. Then list any other rule(s) that you would use, in order. Do not compute the derivative.**

**17.** $f(x) = \sqrt[3]{x\sqrt{x^4 + 2x\sqrt[4]{\dfrac{8}{x + 2}}}}$

6. (a) $f(x) = (x^3 + x - 1)^3$

$f'(x) = 3(x^3 + x - 1)^2 \cdot (3x^2 + 1)$

(b) $f(x) = \sqrt{4x - x^{-1}}$

$f'(x) = \frac{1}{2}(4x - \frac{1}{x}) \cdot (4 + \frac{1}{x^2})$ #

16. (b) $f(x) = (\sqrt{4x^2 + 8} - x^2)^2$

$f'(x) = 2(\sqrt{4x^2 + 8} - x) \cdot (\frac{1}{2}(4x^2 + 8) \cdot (8x) - 2x)$

① Chain Rule
② Product Rule
③ Power Rule
④ Quotient Rule

**In exercises 37 and 38, find the second derivative of each function.**

**37.** (a) $f(x) = \sqrt{x^2 + 4}$ (b) $f(t) = \dfrac{2}{\sqrt{t^2 + 4}}$

**38.** (a) $h(t) = (t^3 + 3)^2$ (b) $g(s) = \dfrac{3}{(s^2 + 1)^2}$

(a) $h'(t) = 2(t^3+3) \cdot (3t^2)$

$h''(t) = 2(3t^2) \cdot (3t^2) + 2(t^3+3) \cdot 6t$

$= 12t^4 + 12t^4 + 36t$

$= 24t^4 + 36t$

§ 2.6

**In exercises 1–18, find the derivative of each function.**

1. $f(x) = 4\sin 3x - x$
2. $f(x) = 4x^2 - 3\tan 2x$
3. $f(t) = \tan^3 2t - \csc^4 3t$
4. $f(t) = t^2 + 2\cos^2 4t$
5. $f(x) = x\cos 5x^2$
6. $f(x) = x^2 \sec 4x$
7. $f(x) = \dfrac{\sin x^2}{x^2}$
8. $f(x) = \dfrac{x^2}{\csc^4 2x}$
9. $f(t) = \sin 3t \sec 3t$
10. $f(t) = \sqrt{\cos 5t \sec 5t}$
11. $f(x) = \dfrac{1}{\sin 4x}$
12. $f(x) = x^2 \sec^2 3x$
13. $f(x) = 2\sin 2x \cos 2x$
14. $f(x) = 4\sin^2 3x + 4\cos^2 3x$
15. $f(x) = \tan\sqrt{x^2+1}$
16. $f(x) = 4x^2 \sin x \sec 3x$
17. $f(x) = \sin^3\left(\cos\sqrt{x^3+2x^2}\right)$
18. $f(x) = \tan^4[\sin^2(x^3+2x)]$

3. $f(t) = \tan^3 2t - \csc^4 3t = (\tan 2t)^3 - (\csc 3t)^4$

$f'(t) = (3\tan^2 2t \cdot \sec^2 2t \cdot 2) + (4\csc^3 3t \cdot (\csc 3t \cot 3t) \cdot 3t)$

① ②

18. $f'(x) = 4\tan^3(\sin^2(x^3+2x)) \cdot \sec^2(\sin^2(x^3+2x)) \cdot 2\sin(x^3+2x) \cdot \cos(x^3+2x) \cdot (3x^2+2)$

③ ④ ⑤

**35.** For $f(x) = \sin 2x$, find $f^{(75)}(x)$ and $f^{(150)}(x)$.

$75 \div 4 = 18 \cdots 3$

$f'(x) = 2\cos 2x$, $f''(x) = -4\sin 2x$

$f'''(x) = -8\cos 2x$, $f^{(4)}(x) = 16\sin 2x$

① $f^{(75)}(x) = -2^{75} \cdot \cos 2x$ ② $f^{(150)}(x) = -2^{150} \cdot \sin 2x$

§2.7

**In exercises 1–4, compute the slope of the tangent line at the given point both explicitly (first solve for $y$ as a function of $x$) and implicitly.**

1. $x^2 + 4y^2 = 8$ at $(2, 1)$
2. $x^3y - 4\sqrt{x} = x^2y$ at $(2, \sqrt{2})$
3. $y - 3x^2y = \cos x$ at $(0, 1)$
4. $y^2 + 2xy + 4 = 0$ at $(-2, 2)$

$3x^2y + x^3y' = 2xy + x^2y'$

2. $(x^3y - 4\sqrt{x})' = (x^2y)'$

$\Rightarrow 3x^2y + x^3 \cdot 1 \cdot y' - \frac{2}{\sqrt{x}} = 2xy + x^2 \cdot 1 \cdot y'$

$\Rightarrow x^3y' - x^2y' = 2xy - 3x^2y + \frac{2}{\sqrt{x}}$

$\Rightarrow (x^3 - x^2)y' = 2xy - 3x^2y + \frac{2}{\sqrt{x}}$

$\Rightarrow y' = \frac{2xy - 3x^2y + \frac{2}{\sqrt{x}}}{x^3 - x^2}$

$y'\big|_{x=2,\, y=\sqrt{2}} = \frac{4\sqrt{2} - 12\sqrt{2} + \frac{2}{\sqrt{2}}}{8 - 4} = \frac{-7\sqrt{2}}{4}$ ($\frac{2}{\sqrt{2}} = \sqrt{2}$)

$y - \sqrt{2} = \frac{-7\sqrt{2}}{4}(x - 2)$